

Chapter 4 exercises — Asymptotic Approximations

76 exercises, extracted from the textbook PDF.

Caveats. (1) Statements are auto-extracted; `fi` / `fl` ligatures dropped by the PDF text layer were repaired by a cleanup pass (a rare odd gap may remain). (2) A $\triangle$ marks exercises whose statement includes a **displayed formula** that did not survive extraction — read those in the book at the cited page. The book PDF is always authoritative.

#	p.	Exercise
4.1	154	Show that $(N/(N+1))^N = O(1)$,
4.2	154	Show that $(1 + 1/N)^N = 1 + O(1/N)$
4.3	154	Show that $N^\alpha = o(N^\beta)$ if $\alpha < \beta$.
4.4	154	Show that, for r fixed, $(N^r)^N = O(N^{r-1} r!)$
4.5	154	Show that $\log N = o(N^\epsilon)$ for all $\epsilon > 0$.
4.6	154	Show that $1 = o(1)$ and $2 + \ln N$
4.7	156	Prove that e is exponentially small for any positive constant ϵ . ($(1/\epsilon)^N$ is, given ϵ , prove that $e^{-N} = O(N^{-M})$ for any fixed $M > 0$.)
4.8	156	Prove that $e^{-\log N}$ and $(\log N)^{-\log N}$ are exponentially small. 2
4.9	156	If $\alpha < \beta$, show that α^N is exponentially small relative to β^N . For $\beta = 1.2$ and $\alpha = 1.1$, find the absolute and relative errors when $\alpha^N + \beta^N$ is approximated by β^N , for $N = 10$ and $N = 100$.
4.10	156	Show that the product of an exponentially small quantity and any polynomial in N is an exponentially small quantity.
4.11	156	Find the most accurate expression for a_n implied by each of the following recurrence relationships: $a_n = 2a_{n/2} + O(n)$, $a_n = 2a_{n/2} + o(n)$, $a_n \sim 2a_{n/2} + n$. In each case assume that $a_{n/2}$ is taken to be shorthand notation for $a_{\lfloor n/2 \rfloor} + O(1)$.
4.12	156	Using the definitions from Chapter 1, find the most accurate expression for a_n implied by each of the following recurrence relationships: $a_n = 2a_{n/2} + O(n)$, $a_n = 2a_{n/2} + \Theta(n)$, $a_n = 2a_{n/2} + \Omega(n)$. In each case assume that $a_{n/2}$ is taken to be shorthand notation for $a_{\lfloor n/2 \rfloor} + O(1)$.
4.13	157	Let $\beta > 1$ and take $f(x) = x^\alpha$ with $\alpha > 0$. If $a(x)$ satisfies the recurrence for $x \geq 1$ with $a(x) = 0$ for $x < 1$
4.14	159	$\triangle$ Use
4.15	159	$\triangle$ Use
4.16	159	$\triangle$ Use
4.17	159	[Miles, cf. Knuth] Show that the polynomial $z^t - z^{t-1} - \dots - z - 1$ has t distinct roots and that exactly one of the roots has modulus greater than 1, for all $t > 1$.
4.18	159	Give an approximate solution for the “ t th-order Fibonacci” recurrence $[t]$

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4.19	159	[Schur] Show that the number of ways to change an N -denomination bill using coin denominations $d_1, d_2, \dots, d_t$ with $d_1 = 1$ is asymptotic to $N^{t-1} \cdot d_1 d_2 \dots d_t (t-1)!$ (See Exercise 3.55.)
4.20	161	Extend Table 4.1 to cover the cases $N = 105$ and 106 .
4.21	163	Expand $\ln(1 - x + x^2)$ as $x \rightarrow 0$, to within $O(x^4)$.
4.22	163	Give an asymptotic expansion for $\ln(N^\alpha + N^\beta)$, where α and β are positive constants with $\alpha > \beta$.
4.23	163	Give an asymptotic expansion for
4.24	163	Estimate the value of $e^{0.1} + \cos(.1) - \ln(.9)$ to within 10^{-4} , without using a calculator.
4.25	163	Show that $1 = 0.000102030405060708091011 \dots 47484950 \dots 9801$ to within 10^{-100} . How many more digits can you predict? Generalize.
4.26	165	Use the nonasymptotic version $\sqrt{}$ formula to give a bound () of Stirling's on the error made in estimating $N! \sim N^{N+1/2} e^{-N}$ with $N \gg N/4$.
4.27	167	Assume that the constant C implied in the O -notation is less than 10 in absolute value. Give specific bounds for H_{1000} implied by the absolute formula $H_N = \ln N + \gamma + O(1/N)$ and by the relative formula $H_N = \ln N (1 + O(1/\log N))$.
4.28	167	Assume that the constant C implied in the O -notation is less than 10 in absolute value. Give specific bounds for the 10th Catalan number implied by the relative formula $() (1/N) \sim \frac{1}{2N} \sqrt{4N - 3}$.
4.29	167	Suppose that $f(N)$ admits a convergent representation $\sum_{k \geq 0} f(N) = a_k N^{-k}$
4.30	167	Construct a function $f(N)$ such that $f(N) \sim$
4.31	170	Prove or disprove the following, for $N \rightarrow \infty$: $() e^N = O(N^2)$,
4.32	170	Give an asymptotic expansion for $e^{1/(N+1)}$ to within $O(N^{-3})$.
4.33	171	Calculate $(H_N)^2$ to within $O(1/N)$, then to within $o(1/N)$.
4.34	172	Derive an asymptotic expansion for $\cot x$ to $O(x^4)$.
4.35	172	Derive an asymptotic expansion for $x/(e^x - 1)$ to $O(x^5)$.
4.36	173	Carry out the expansion for the Catalan numbers to within $O(N^{-4})$ accuracy. ()

#	p.	Exercise
4.37	173	Calculate an asymptotic expansion for $3N N/(N+1)$.
4.38	173	Calculate an asymptotic expansion for $(3N)!/(N!)^3$.
4.39	174	What is the approximate value of $1 - ? N (\ln N) N$
4.40	174	Give a three-term asymptotic expansion of $1 - . N$
4.41	174	Suppose that interest on a bank account is “compounded daily”—that is, $1/365$ of the interest is added to the account each day, for 365 days. How much more interest is paid in a year on an account with \$10,000, at a 10% interest rate compounded daily, as opposed to the \$1000 that would be paid if interest were paid once a year?
4.42	174	Simplify the asymptotic expression $\exp\{1+1/N + O(1/N^2)\}$ without losing asymptotic accuracy. ()
4.43	174	Find an asymptotic estimate for $\ln(\sin((N!) - 1))$ to within $O(1/N^2)$.
4.44	174	Show that $\sin(\tan(1/N)) \sim 1/N$ and $\tan(\sin(1/N)) \sim 1/N$. find the order of growth of $\sin(\tan(1/N)) - \tan(\sin(1/N))$.
4.45	174	Find an estimate for HT_N , where T_N is the N th Catalan (asymptotic) number, to within $O(1/N)$.
4.46	175	Let a_n be defined as the unique positive root of the equation $n = a_n \tan(a_n)$ for $n > 1$. Find an asymptotic estimate for a_n , to within $O(1/(\log n)^3)$.
4.47	175	Give the reversion of the power series $y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + O(x^4)$. (Hint : Take $z = (y - c_0)/c_1$.)
4.48	177	Give an asymptotic estimate for
4.49	177	Give an asymptotic estimate for
4.50	177	Give an asymptotic estimate for
4.51	177	Give an asymptotic estimate for
4.53	182	Develop an asymptotic expansion for $\sum_{0 \leq k \leq N} 1/(1 + k/N)$
4.54	182	Show that $\sum_{0 \leq k \leq N}$
4.55	186	Evaluate γ to 10 decimal places.
4.56	186	Show that the generalized (second-order) harmonic numbers admit the asymptotic expansion $\sum$
4.57	186	Derive an asymptotic expansion for $\sum$

#	p.	Exercise
4.58	186	Use Euler-Maclaurin summation to estimate $\sum \sqrt{k}$, $1 \leq k \leq N$
4.59	186	Derive full asymptotic expansions for $\sum (-1)^k k$
4.60	186	$\triangle$
4.61	192	$\triangle$ Prove that
4.62	192	Use Stirling's formula for $\ln N!$ to prove the relative bounds for the Ramanujan Q- and R-distributions given in eorem 4.4 and eorem 4.5, respectively.
4.63	194	Write a program to compute exact values of the binomial distribution to single-precision floating point accuracy.
4.64	198	Carry out the normal approximation to $O(1/N^2)$ for the case $k = \sqrt{N} + O(1)$.
4.65	198	Plot the smallest k for which the binomial probabilities are greater than .001 as a function of N .
4.66	202	$\triangle$ Give an asymptotic approximation to
4.67	202	Give an asymptotic approximation to N for p fixed. $k p (1 - p)$ (Hint : Shift so that the largest terms in the distribution are at $k = 0$.)
4.68	202	Give an asymptotic approximation of the binomial distribution for the case where $p = \lambda / N$.
4.69	202	Give an asymptotic approximation of the binomial distribution for the case where $p = \lambda / \ln N$.
4.70	205	By applying Euler-Maclaurin summation to the functions $x e^{-x/2}$ and $3 x^3 e^{-x/2}$, show that 2
4.71	207	Show that $P(N) =$
4.72	207	Find a direct argument showing that $R(N) - Q(N) \sim 2/3$.
4.73	211	Find an asymptotic estimate for
4.74	211	Find an asymptotic estimate for
4.75	211	Find an asymptotic estimate for
4.76	214	Find an asymptotic estimate for
4.77	214	Find an asymptotic estimate for